Joint Signing-Partner and Audit-Firm Tenure and Environmental Disclosure: Evidence from Completing Partner Spells

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Primary sample: 31,274 firm-years across 4,166 firms, 2013-2023

Data: CSMAR audit database (commercially licensed) and Huazheng ESG pillar scores (public via Mendeley Data, Lin 2025)

Companion document: Data Availability and Source File Note

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Abstract

China has capped a signing partner's service to one client at five consecutive years since 2004; the 2022 repeal removed one instrument, but the CICPA Code and 2023 Measures continued the obligation, so every pair knew its end date years ahead. We ask whether environmental disclosure moves along that dated clock, using the Huazheng environmental pillar (E) on 31,274 firm-years and 4,166 firms (2013-2023). A mandatory partner change has no average association with E (-0.075, p = 0.407); the tenure path is non-monotonic, rising into year four and falling in year five.

Among completing-spell firms (spells ending inside the window), the fall concentrates where the partner clock, carrying the deadline, and the audit-firm clock, carrying none at five, both stand at four or five: net of pair-by-firm fixed effects the contrast is +0.653 (t = 5.66; bootstrap p < 0.001) at five and, in a post hoc widening reported as descriptive, +0.622 (t = 6.13) across the window, against +0.057 (t = 1.02) outside it. The +0.653 is a differential between pair-firm clusters containing a joint five-year cell and other pair-firm clusters, not a cell-level both-clocks-at-five effect. The design is associational throughout; Section 4.3's graded firm-clock profile is consistent with an anticipated date but does not establish one. Reweighting the joint firms' comparators on firm means, persistence and spell completion moves the joint-minus-below gap from +1.088 to +0.673 (t = 2.21), at a cost of more than four-fifths of the comparators, and clock-unrelated turnover is bounded at 0.30 and 0.42 points, coefficients near zero (+0.003 and +0.096). The pattern is not detected among firms observed in all eleven years (-0.11, t = -0.89; MDE 0.36; +0.044, t = 0.19, in their joint cells); balancing persistent against non-persistent firms on ten firm means cuts their gap from 1.310 to 0.514 (t = 1.40; Table 10, row 9), so composition narrows, not removes, that null.

Keywords: audit partner rotation; auditor tenure; environmental disclosure; ESG ratings; joint tenure; statutory deadlines

1. Introduction

Most research studies either the rotation event or the level of tenure (Chi, Huang, Liao, and Xie 2009; Firth, Rui, and Wu 2012; Gipper, Hail, and Leuz 2021). We think the event is the wrong margin: rotation is anticipated, its timing follows when each pair started, and the pairs that rotate are selected. What the rule gives us is a known future date, and a known date gives both sides reason to act before it arrives.

Environmental disclosure is where the stakes show. Environmental claims are soft, forward-looking and rarely audited, so an audit team can quietly accommodate or tighten them (Lyon and Maxwell 2011; Kim and Lyon 2015; Delmas and Burbano 2011); because audited reporting and voluntary disclosure are complements (Ball, Jayaraman, and Shivakumar 2012), that tolerance shows up in what the client says.

Two forces run along the clock. Early on the partner knows the client poorly and checks hard; as tenure builds, challenging one more soft claim costs more. That pushes the path up. The cap disciplines the last year: a team that must leave sees it coming, and a client about to lose a familiar partner may adjust first. That pushes year five down.

The audit-firm clock makes the central point. A firm's tenure at a client has no statutory deadline at five years, but the two clocks meet often enough to test: 1,764 of the 3,223 sample firms that reach a partner-clock-five year inside the window (54.7 percent; 42.3 percent of all 4,166 firms, 943 of which have no clock-five year in the window) sit in the joint cell in their first partner-clock-five year, and joint cells are 53.4 percent of clock-five firm-years among firms whose spells complete inside the window against 21.9 percent among firms we observe in all eleven years (Table 6A, Panel B; Note Section 8.17). If the deadline force is real, the contraction should be sharpest where the two clocks meet at four or five - a coincidence point with no statutory content.

Three results follow, and the design behind them is associational throughout: no untreated group, no variation in treatment timing, and no difference-in-differences logic anywhere, including our reading of the 2022 repeal. First, the event margin is empty on average: a mandatory partner change has no average association with E (-0.075, p = 0.407), although the event study finds a rotation-year dip (-0.541, t = -3.56) in cells too underpowered to carry the result (Section 4.11). Second, the path is non-monotonic and the fall is a joint-clock event: +0.653 (t = 5.66) where both clocks reach five, +0.622 (t = 6.13) across the four-or-five high-tenure window, against +0.057 (t = 1.02) where the firm clock is outside it; the raw fall of 0.646 is an upper bound that pair-by-firm fixed effects cut to 0.295. Third, the pattern is graded along the firm clock - significant at four, largest at five, absent at six and seven - survives entropy balancing at the firm and cluster levels and spell-level matching (Appendices A.8, A.12 and A.13), and vanishes among firms we can watch for all eleven years (-0.11, t = -0.89; +0.044, t = 0.19, in their joint cells).

We add to three literatures. For partner rotation, the action sits in the final service years, and only where the audit-firm clock is in high tenure too. For tenure and disclosure, our path is non-monotonic and pillar-specific, which no linear tenure

framing predicts (Gul, Fung, and Jaggi 2009; Gul, Wu, and Yang 2013; Lin, Shen, Wang, and Yu 2024). For ESG ratings, a statutory audit clock moves the hardest-to-verify pillar along a dated path, in a literature where vendors disagree in level (Berg, Kolbel, and Rigobon 2022; Christensen, Serafeim, and Sikochi 2022; Krueger, Sautner, Tang, and Zhong 2024).

2. Institutional Background and Hypotheses

2.1 The rule, the two clocks, and the repeal

The CSRC and the Ministry of Finance capped a signing partner's consecutive service at five years with a two-year cooling-off period (Zheng Jian Kuai Ji Zi [2003] No. 13, effective January 1, 2004). Service years accumulate across audit-firm moves at the same client (Article 4), so a partner who switches firms keeps the same clock; when both partners hit the cap together, one may stay one more year (Article 7), a provision about the two partners, not the firm.

CSRC Announcement [2022] No. 42 repealed the rule in August 2022, but the obligation did not lapse. The Chinese Institute of Certified Public Accountants (CICPA) Code of Ethics (2020) already capped the engagement and signing partner at five consecutive years, and the joint Measures on the Engagement of Accounting Firms by State-Owned and Listed Companies (Ministry of Finance, SASAC, and CSRC 2023, Article 13) restated the five-year cap for listed companies, kept the rule that service years accumulate across audit-firm moves, and set the cooling-off period at five consecutive years; Independence Standard No. 1 (Ministry of Finance 2024) restated it again and treats the second signing partner as an engagement partner. What changed in 2022 was the instrument, not the clock (Section 4.9). The audit-firm clock carries no statutory content at five years either: the 2023 Measures cap audit-firm tenure at eight years, extendable to ten, and only for state-owned enterprises (Article 12), which are 29.4 percent of our firms. The five-year coincidence we study is therefore not a regulatory construct on either clock.

Two clocks run on one engagement. The **partner clock** counts the longer-serving partner's consecutive years; it carries the statutory deadline at five. The **audit-firm clock** counts the firm's consecutive years at the client; it has no deadline. The two are only loosely related - the correlation of the counts is 0.22.

2.2 Why the two clocks might meet

Familiarity against deadline discipline. Early in a partnership the team contests soft claims readily; as it matures, contesting costs more, while the rating rewards saying more about hard-to-verify topics. The offset is the cap: a team that must leave anticipates leaving, and a client about to lose a familiar partner may reposition first. We hypothesise that familiarity pushes the path up and the deadline pushes year five down; because the cap binds at five, any fall belongs in year five.

What the firm clock brings. The firm clock has no deadline, but China's independence rules attach review machinery to long associations. The CICPA Code (2020, Chapter 11) requires a firm to assess the familiarity threat on long-serving teams and apply safeguards - rotating people off the team, independent review, periodic quality review - and Independence Standard No. 1 (2024, Article 86) adds an external second review at ten years. Nothing is pinned to five years, so we do not claim the firm clock carries a deadline.

The testable claim. If the deadline force operates, the contraction should be sharpest where the statutory partner date and a five-year firm anniversary coincide. We have no structural model of why that point matters, so our hypothesis is about where the concentration sits, not why. Pairs that survive both clocks to year five are the more stable ones, so the joint cell is selected (Sections 4.4 and 4.13).

Pillar specificity. If the fall is about disclosure rather than tenure in general, it should live in the pillar whose claims are hardest to verify.

2.3 Hypotheses

Table 1. Hypotheses.

No.	Statement
H1	E is flat through years two and three, rises in year four, and falls in year five.
H2	The year-four-to-year-five contrast appears in the firm-and-year specification, as a descriptive upper bound.
H3	The path is specific to the environmental pillar; S and G do not show H1-H2.
H4	The fall is larger where mandates and political evaluation leave less room for soft claims - that is, in non-state firms (Wang, Wong, and Xia 2008).
H5	**The fall is concentrated where both the partner clock and the audit-firm clock are in high tenure, and the profile is graded across firm-clock years four, five, six and seven rather than unique to five; Section 4.3 reports the coefficients behind this hypothesis.**

H5 is the title claim; H1-H4 describe the raw partner-clock path.

3. Data, Variables, and Identification

3.1 Data and sample

Audit records come from the CSMAR audit database (1997-2025), as do firm financials, executive covariates and environmental-penalty records. ESG pillar scores are the Huazheng rating file, public via Mendeley Data (Lin 2025). Our partner clock is the consecutive-service count of the longer-serving signing partner, rebuilt from the name sequence; over the full 2003-2025 name panel it matches the within-firm individual clock in all 32,207 firm-years with a name record (one 2018 firm-year has none), and dropping that firm-year leaves the fall unchanged (+0.646, t = 7.44). The window is 2013-2023.

Selection has two branches, because event designs and clock-keyed designs need different fields to be complete. Table 2 gives both.

Table 2. Sample funnel, two branches (one clock construction, two completeness requirements)

Branch	Step	Rows	Firms
Common	CSMAR audit records 1997-2025	88,851	6,117
Common	Restrict to 2013-2023	52,558	5,619
Common	Match to Huazheng pillar scores	33,923	4,777
A: event designs	Exclude financial and ST firm-years	32,608	4,586
A	Require rotation classification and clock	32,537	4,586
A	Require complete controls - **event-design sample**	**32,017**	**4,571**
B: clock-keyed	Retain rows with missing ST/financial flags and missing rotation classification, provided clock and controls are complete	32,755	4,672
B	Require clock and controls - **clock-keyed sample**	**32,208**	**4,666**
B	Tenure one to five	31,734	4,626
B	Singleton absorption - **primary firm-year sample**	**31,274**	**4,166**

Branch B adds 191 firm-years to Branch A: 126 with missing ST or financial flags and 65 without the rotation classification, all with a complete clock and controls. The pair-by-firm designs put back the 126, so pair-by-firm cells are not a subset of Branch A.

3.2 Two estimands, kept apart

We report two estimands and never mix them. Sign convention: every year-four-to-year-five contrast below is year four minus year five, so a positive number is a decline in E; Tables 7, 11 and 12 and the penalty contrasts follow the same rule. Tables that report coefficients on E itself - the event-study cells of Tables 13 and A.3-

A.4 and the shock contrasts of Section 4.12 - use the opposite sign, and each carries its own note.

1. **The reporting estimand (descriptive, an upper bound).** The year-four-to-year-five contrast in the firm-and-year fixed-effects specification on the 31,274-firm-year primary sample - the raw partner-clock fall.
2. **The title estimand (primary).** The interaction-implied joint-cluster differential in the pair-by-firm specification at the five-year point, where both clocks read five (+0.653, t = 5.66; Section 4.3); its post hoc widening to the four-or-five high-tenure window, +0.622 (t = 6.13), is reported as descriptive (Appendix A.12). It is a joint-cluster differential: pair-by-firm fixed effects absorb all time-invariant match quality between a pair and a client, and we estimate it as the main effect of the two-way-demeaned contrast plus its interaction with a joint-cluster indicator, which keeps one fixed-effect structure across all 7,751 clusters. It compares clusters, not destination cells. Two numbers in this paper are often read as the title estimand and are not the title estimand: the cell-level destination estimate of +1.192 (Table 6, 1,764 firms falling in the joint cell) and the joint-subsample estimate of +1.201 (Table 8, 970 clusters, fixed effects re-absorbed within joint clusters only). Neither is the +0.653 estimand, and the three are not alternative estimates of one quantity.

Table 2A. Estimand map: the quantities quoted in this paper and how they differ. All rows are year-four-minus-year-five contrasts in the sign convention of this section (positive = decline); inference on pair-by-firm rows is the restricted wild-cluster bootstrap at 9,999 draws, and MDE is at 80 percent power.

Quantity (table; unit)	Coef.	SE	t	WCR p	MDE	N / clusters	Status
Raw firm-year fall (Table 4; firm and year FE)	0.646	0.087	7.44	n/a	0.243	31,274 / 4,166 firms	Pre-specified
Pooled pair-by-firm fall (Table 8; all clusters)	+0.295	0.130	2.27	0.029	0.36	19,376 cells / 7,751 clusters	Pre-specified
Title: implied joint effect at five (Table 8)	+0.653	0.115	5.66	< 0.001	0.32	19,376 cells / 7,751 clusters	Pre-specified (title)
Four-or-five window widening (Table 8; A.12)	+0.622	0.102	6.13	< 0.001	0.28	1,108 cells / 1,105 clusters	Post hoc, descriptive
Destination-cell fall (Table 6; joint-cell firms)	+1.192	0.135	8.85	n/a	0.38	1,764 firms	Corroborating
Joint-subsample fall (Table 8; joint clusters)	+1.201	0.289	4.15	< 0.001	0.81	2,622 cells / 970 clusters	Corroborating

Rows 3, 5 and 6 are not alternative estimates of one quantity: row 3 is a joint-cluster differential estimated under one fixed-effect structure, while rows 5 and 6 sit on different units and samples (Section 4.3). Row 4 is a descriptive widening of row 3, pooled after seeing the firm-clock profile

(Appendix A.12).

Inference uses the restricted (null-imposed) Rademacher wild-cluster bootstrap at 9,999 draws for pair-by-firm-clustered estimates (MacKinnon and Webb 2017; Webb 2023), and firm-clustered conventional inference with a wild bootstrap where noted (Cameron, Gelbach, and Miller 2008). We report minimum detectable effects at 80 percent power (2.80 x standard error).

3.3 What the design can and cannot do

We estimate within-firm life-cycle contrasts: tenure dummies, firm and year fixed effects, and controls. There is no untreated group and no variation in timing, so no difference-in-differences logic applies. We address three threats directly: time-invariant match quality, by pair-by-firm fixed effects; composition, by a balanced panel of firms present in all eleven years (Section 4.4); and clock misclassification at the cap boundary, by excluding firm-change spells, name-matched incoming service, and non-conforming transitions; the clock starts when a firm enters the audit panel, so service rendered before listing, which the 2023 Measures would fold into the count (Article 13), is not observed and tenure at the cap may be understated for recently listed firms. What we cannot rule out is any shock moving with spell age within a firm-year; every contrast here is associational.

4. Results

4.1 The rotation event is null

Table 3 reports the event design (firm and year FE, controls, firm-clustered, N = 32,017). A mandatory partner change has no average association with E (-0.075, p = 0.407); neither does any partner change (-0.067, p = 0.314), nor a full change out of a clock-five year (1,479 changes, +0.003, p = 0.98). The event margin is empty, which is why we turn to the path.

Table 3. Rotation-event design (firm and year FE, controls, firm-clustered; N = 32,017)

Specification	Coefficient	p
Mandatory change	-0.075	0.407
Any partner change	-0.067	0.314
Full change out of a clock-five year (1,479 changes)	+0.003	0.98

4.2 The tenure path is not monotonic

Table 4 reports the firm-and-year specification on the primary sample. The path is flat through year three, rises in year four, and falls in year five. The year-four-to-year-five contrast is 0.646 (SE 0.087, t = 7.44; bootstrap p < 0.001): 0.09 of the E standard deviation (7.32 points) and about 36 percent of the median within-firm year-over-year change (1.79 points). Call this the raw upper bound - the contrast before absorbing any time-invariant match quality.

Table 4. Tenure path, primary sample (firm and year FE, full controls, firm-clustered; N = 31,274, 4,166 clusters)

Quantity	Coefficient	SE	t
Year 2 (vs. year 1)	+0.071	0.080	0.89
Year 3	+0.084	0.097	0.86
Year 4	+0.490	0.108	4.53
Year 5	-0.156	0.111	-1.40
Year-four-to-year-five contrast	**+0.646**	**0.087**	**7.44**

The design detects 0.243 points at 80 percent power, so the year-two and year-three nulls are informative. The fall holds under industry-by-year fixed effects and grows when firm-change spells are dropped (Table 12).

4.3 The fall sits where both clocks are in high tenure

Table 5 enters both clocks together. The year-four elevation belongs to the partner clock; the year-five level is flat on the partner clock alone, negative on the firm clock alone and most negative where both clocks reach five (-0.593, t = -3.85; Table 6's counterpart is -0.547, t = -4.08).

Table 5. Both clocks entered together (firm and year FE, full controls, firm-clustered; N = 31,274; the joint cell is the mutually exclusive four-cell version of the same specification, and the first five rows are coefficients on E levels while the last two rows are year-four-minus-year-five declines, where positive means a decline.

Quantity	Coefficient	SE	t
Partner clock, year 4	+0.445	0.111	4.02
Partner clock, year 5	-0.024	0.118	-0.21
Audit-firm clock, year 4	+0.193	0.118	1.63
Audit-firm clock, year 5	-0.381	0.109	-3.50
Both clocks at five (four-cell specification)	-0.593	0.154	-3.85
Four-to-five contrast, partner clock	+0.469	0.091	5.14
Four-to-five contrast, firm clock	+0.573	0.114	5.02

Table 6 assigns each firm to the cell its firm clock occupies in the firm's first clock-five year. The fall column is the one that matters: +1.192 (t = 8.85) in the joint cell, against +0.104 (t = 0.52) below five and +0.191 (t = 1.32) above. Firms with no clock-five year inside the window enter only through their year-four cells. Combining the joint cell with the 128 firms whose firm clock reads four, the fall is +1.114 (SE 0.126, t = 8.85) on 1,892 firms (Appendix A.12).

Table 6. The two clocks at the partner clock's five-year point (firm and year FE, full controls, firm-clustered; N = 31,274)

Cell (partner clock at five)	Firms	Year-five level	t	Year-four-to-year-five fall	t
Audit-firm clock also at five (joint)	**1,764**	**-0.547**	**-4.08**	**+1.192**	**8.85**
Audit-firm clock below five	463	-0.120	-0.56	+0.104	0.52
Audit-firm clock above five	996	+0.221	1.32	+0.191	1.32
No clock-five year inside the window	943	-	-	-	-

Table 6A gives the joint distribution behind those cells. The coincidence is common but far from universal: 1,764 of the 3,223 firms that reach a partner-clock-five year inside the window sit at the audit-firm clock's fifth year in their first partner-clock-five year (54.7 percent; 42.3 percent of all 4,166 firms, 943 of which never reach one) and 128 more sit at four, while 943 never reach a partner-clock-five year inside the window. Panel B is why the balanced-panel null of Section 4.4 is uninformative rather than contradictory: joint cells are 53.4 percent of clock-five rows among firms whose spells complete inside the window but only 21.9 percent among firms observed in all eleven years, so the persistent firms have few joint rows to lose. The firm-year counterpart of Panel A is 2,018 joint firm-years among the 31,274 rows of the estimation sample (Note Section 8.16).

Table 6A. Joint distribution of the two clocks. Panel A: all 4,166 primary-sample firms by the audit-firm clock reading in the firm's first partner-clock-five year (firms with several clock-five years use the first; firms with none are shown on their own row); clocks two and three are not reported separately in the deposited placebo cells and are combined here. Panel B: the share of partner-clock-five firm-years that are joint, by panel persistence (Note Section 8.17).

Panel A: firms by the audit-firm clock reading in the first partner-clock-five year

Audit-firm clock	Firms
1	101
2 and 3	234
4	128
5 - joint	1,764
6	173
7	157
8 to 19	666
No partner-clock-five year inside the window	943
Total	4,166

Panel B: joint share of clock-five firm-years, by panel persistence

Sample	Joint share
Firms whose partner spells complete inside the window	53.4 percent
Firms observed in all eleven years	21.9 percent

Table 7 repeats the construction at every other tenure year; only the five-year point moves, and year six is thin (235 firms, MDE 1.82).

Table 7. Joint placebo: the year-(t-1)-to-year-t fall among firms with both clocks at t (firm and year FE, controls, firm-clustered; positive = decline, the convention used throughout)

Tenure year t	Contrast	Fall	t	Firms	MDE (80%)
2	Year 1 to year 2	-0.001	-0.01	1,716	0.30
3	Year 2 to year 3	+0.081	0.70	1,431	0.32
4	Year 3 to year 4	-0.130	-1.04	1,885	0.35
5	**Year 4 to year 5**	**+0.938**	**6.81**	**2,066**	**0.39**
6	Year 5 to year 6	-0.043	-0.07	235	1.82

Entropy balancing (Appendix A.8) reweights the other 2,402 firms onto the 1,764 joint firms: the joint fall moves from +1.192 to +1.182 (t = 8.79) and the joint-minus-firm-clock-below gap narrows only from +1.088 to +0.972. Three exploratory splits (Appendix A.9) find no larger fall where firm review machinery should bind harder.

Appendix A.12 asks how far past the joint point the pattern reaches. At the pair-by-firm margin the profile is graded rather than knife-edged: firm clock four returns +0.417 (t = 2.82), about two-thirds of the joint estimate, firm clock five +0.653 (t = 5.66), six +0.056 and seven -0.216. A deadline known years ahead should not be expected to bite in one year only, and the four-year reading is consistent with adjustment beginning in the penultimate year; we report it rather than build on it. The window is post hoc: our pre-specified estimand is the five-year point, and we pooled four and five only after seeing this profile, so the window estimate is descriptive and we have no institutional evidence that review cycles begin at four. Pooling those two years returns +0.622 (t = 6.13; 1,105 clusters). One point of interpretation bears on the title. The +0.653 contrast is the year-four-minus-year-five contrast interacted with a cluster-level indicator for pair-by-firm clusters containing at least one joint five-year cell, so it is a joint-cluster differential, not a cell-level both-clocks-at-five effect; a joint cluster also contains non-joint cells. The cell-level quantities are the destination-cell estimate (+1.192, Table 6) and the joint-subsample estimate (+1.201, Table 8), which are on different units and samples and are not substitutes for +0.653. Section 5 of the companion Note records the same estimand definition. Cluster-level entropy balancing leaves the joint-minus-other gap at +0.608 against +0.596 unweighted. Across the four firm-clock thresholds of Appendix A.12 - clock four, five, six and seven - the Bonferroni threshold is |t| = 2.50 (0.05 over four tests), and only the clock-four and clock-five readings clear it; the post hoc four-or-five window is a fifth comparison we report as descriptive and do not count in that family.

Table 6's +1.192 (destination cells, 1,764 firms) and Table 8's +1.201 (pair-by-firm clusters, 970) are corroborating estimates on different units, not the same quantity. Table 8 reports the title estimand at its five-year point: the interaction-implied joint effect is +0.653 (t = 5.66; bootstrap p < 0.001) against +0.057 (t = 1.02) where the firm clock sits outside the high-tenure window, and the design detects 0.32 points there. On the joint subsample alone the fall is 1.201 (t = 4.15); estimated without the interaction over all 7,751 clusters it is 0.295 (t = 2.27), weighted by identifying variation rather than cell count.

Table 8. Joint high-tenure contrast, pair-by-firm specification (pair-by-firm and year FE, controls, pair-by-firm-clustered; restricted wild-cluster bootstrap at 9,999 draws; MDE at 80 percent power). The four-or-five window row is estimated on the 1,105 clusters whose firm clock reaches four or five and, within them, on their firm-clock-four-or-five cells only, which is why its cell count is 1,108 rather than 19,376; every other row uses the full 7,751 clusters. The window row is a post hoc widening of the five-year estimand and is reported as descriptive (Appendix A.12).

Specification	Fall	SE	t	WCR p	MDE	Clusters	Cells
Five-year point: implied joint effect (D2 + D2 x joint)	**+0.653**	**0.115**	**5.66**	**< 0.001**	**0.32**	7,751	19,376
High-tenure window (firm clock 4 or 5): implied joint effect	**+0.622**	**0.102**	**6.13**	**< 0.001**	**0.28**	1,105	1,108
D2 x joint (five-year point)	+0.596	0.129	4.63	< 0.001	0.36	7,751	19,376
Non-joint main effect (D2)	+0.057	0.056	1.02	0.30	0.16	7,751	19,376
Joint subsample only	+1.201	0.289	4.15	< 0.001	0.81	970	2,622
All other clusters	-0.161	0.148	-1.09	0.27	0.41	6,781	16,754
All clusters, no interaction	+0.295	0.130	2.27	0.029	0.36	7,751	19,376

Dropping the ten largest pair-firm clusters leaves the all-cluster fall at 0.266 (t = 2.11; bootstrap p = 0.032). Table 9 splits joint clusters by pillar and by ownership.

Table 9. Joint-level heterogeneity (pair-by-firm and year FE, controls, pair-by-firm-clustered)

Panel A: pillar split

Split	E	S	G
Joint five-year clusters (970 clusters, 2,622 cells)	**+1.201 (t 4.15)**	-0.224 (t -0.67)	-0.103 (t -0.41)
All other clusters (6,781 clusters, 16,754 cells)	-0.161 (t -1.09)	+0.272 (t 1.32)	+0.079 (t 0.54)

Panel B: ownership split within joint clusters

Ownership split, joint clusters	**Fall**	**t**
Non-state (1,668 cells)	+1.463	3.92
State (954 cells)	+0.750	1.55

4.4 The pattern does not survive among persistent firms

Does the path hold for firms we can watch the whole time? No. Among the 1,203 firms present in all eleven years, on the same tenures-one-to-five window (13,122 firm-years), the fall is -0.113 (t = -0.89; MDE 0.355). Table 10 collects the variants; all agree.

Table 10. Composition diagnostics (firm and year FE, controls, firm-clustered unless noted; tenures one-to-five; MDE at 80 percent power). Rows 5 and 6 are implied contrasts from one regression on all 31,274 firm-years, not separate subsample estimates. Row 9 reports the gap between the two falls rather than a fall: non-balanced minus balanced, estimated under entropy-balancing weights on ten firm means that leave 928 effective comparison firms (Data Note, Section 8.17); its MDE is 2.80 times its standard error.

Sample	Fall	t	N	MDE
Balanced panel (firms observed all eleven years)	-0.113	-0.89	13,122	0.36
Balanced panel, firms that reach clock five	-0.098	-0.77	11,321	0.36
Firms whose observed span touches both window edges (2013 and 2023)	-0.045	-0.42	18,601	0.30
Firms with at least ten observed years	+0.021	0.20	18,442	0.31
Non-balanced firms (interaction specification)	+1.198	10.19	31,274	0.33
Balanced firms (interaction specification)	-0.112	-0.88	31,274	0.36
Balanced panel, joint clusters only (pair-by-firm FE; 271 clusters)	-0.096	-0.23	897 cells	1.17
Non-balanced firms matched to balanced on size, ownership, listing age and leverage (coarsened exact matching)	+0.934	6.56	25,683	0.40
Balanced and non-balanced firms balanced on ten firm means (entropy balancing: gap, not a fall)	0.514	1.40	31,274	1.03

Three checks show the difference is not empty cells. The balanced panel is not short of cap observations (2,622 year-four, 1,887 year-five and 413 joint cells; 86.9 percent of its firms reach clock five), and the pattern is reshaped rather than absent: the interaction implies +1.198 (t = 10.19) among non-balanced firms against -0.112 (t = -0.88) among balanced ones.

Balanced firms are not a random subset: they are larger (mean size 22.6 against 22.1), older and more often state-controlled (36.9 against 29.4 percent). Composition carries about a fifth of the gap. Matching exactly on size, ownership, listing age and leverage (2,721 firms, 25,683 rows) leaves the non-balanced fall at +0.934 (t = 6.56) against -0.118 (t = -0.93) among balanced firms, a gap of 1.052 (t = 5.52) against 1.310. Balancing on all ten firm means cuts the gap to 0.514 (t = 1.40; the last row of Table 10, with its effective control firms and minimum detectable effect reported there) - the largest single reduction in the table - but it does so by weighting on listing age, which is close to what membership in the group means, and it costs more than half the comparison group: effective control firms fall from 2,402 to 928. This ten-mean battery (treated: the 1,203 balanced firms; covariates: the ten controls of Section 3.1) is a different exercise from the joint-firm reweighting of Appendix A.13 (treated: the 1,764 joint firms; constraint sets of nine means, plus persistence, plus completion), whose effective comparison firms run 1,634, 1,629 and 428; the two batteries share no row. We therefore read the composition checks as narrowing the selection concern rather than removing it: the gap survives coarsened exact matching at 1.052 (t = 5.52) and falls to insignificance only under a reweighting that concentrates on the variable that defines the two groups. Two rows of Table 10 that sound alike are not the same group: firms observed at both window edges are long-lived and behave like the balanced panel, whereas non-balanced firms are those whose partner spells finish inside the window.

The firm-level counterpart of the cluster reweighting of Section 4.10 asks the same question one

level down (Tables A.7 and A.8). Joint firms are not a random draw from the window: against the remaining 2,402 firms they are observed in 6.58 years on average against 8.18, only 15.8 percent of them appear in all eleven years against 38.5 percent of the rest, and they are younger (-0.63 standard deviations, the largest of the twelve covariate differences) and smaller (-0.31). Entropy balancing on the nine firm means of Appendix A.8 reproduces that appendix's estimate (+1.182); adding the persistence descriptors, years observed and years covered, to the constraint set changes nothing (joint fall +1.182, t = 8.78; gap +0.993, t = 3.70). The firm-level analogue of the cluster-level composition constraint - whether the firm's own partner spell reaches clock five inside the window, which is one by construction for joint firms and 0.61 for the rest - drives 943 of the 2,402 controls to zero weight and leaves 428 effective ones; even then the joint fall is +1.177 (t = 8.74) and the gap +0.673 (t = 2.21). Reweighting narrows the composition reading at the firm level, as it did at the cluster level, without removing it. What removes the cell-level pattern is persistence itself: among the 1,203 firms observed in all eleven years the joint-cell fall is +0.044 (t = 0.19; MDE 0.66) against +1.469 (t = 9.16) among the rest, and the pair-level row of Table 10 says the same at the cluster level. The joint-minus-other differential survives every firm-level reweighting we can construct, but it is identified off firms whose spells finish inside the window.

A pre-trend check narrows the selection reading. Extending the interaction to years two and three gives jointly zero pre-deadline coefficients (Wald chi-square(2) = 1.38, p = 0.50) and an implied year-two-to-three difference of -0.056 (t = -0.37; MDE 0.42): the groups do not diverge before the deadline can bite. Divergence starts one step later: -0.567 (t = -3.51) at the year-three-to-four step and +1.310 (t = 7.56) at the year-four-to-five step, as a deadline reading predicts and a difference-in-levels reading does not.

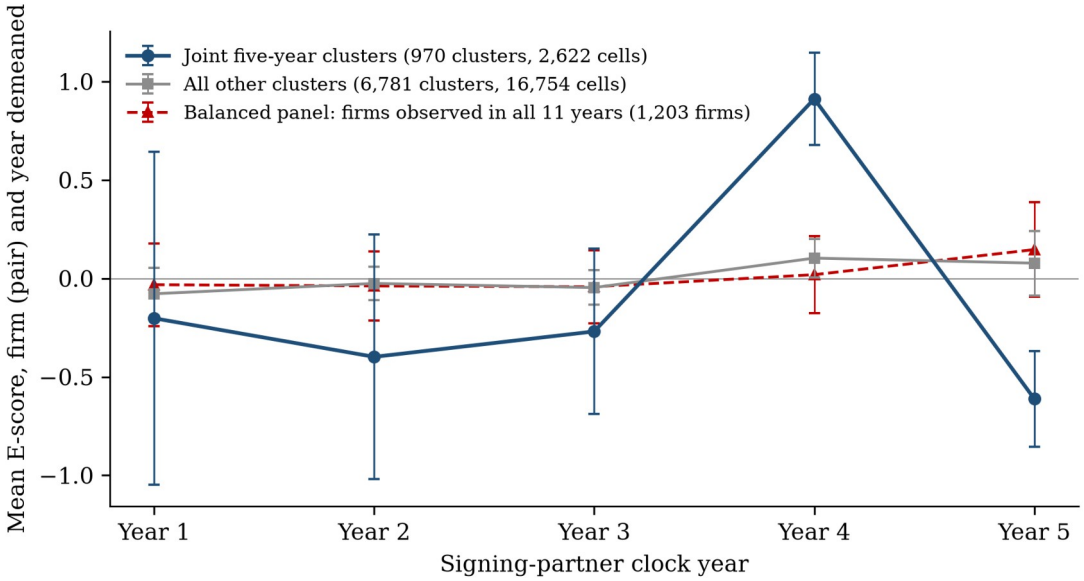


Figure 1. Mean E-score by signing-partner clock year, residualized on pair-firm (or firm) and year fixed effects. Joint five-year clusters (970 clusters, 2,622 cells); all other clusters (6,781 clusters, 16,754 cells); balanced panel (1,203 firms). Whiskers are 95 percent confidence intervals of the cell means, computed from cell-level standard errors that the deposited code regenerates; the archived file records the means and counts (Note Section 8.17). Plotted values are residualized cell means, not regression estimates; we quote the regression contrasts of Tables 8 and 12 (+1.201, -0.161 and -0.113).

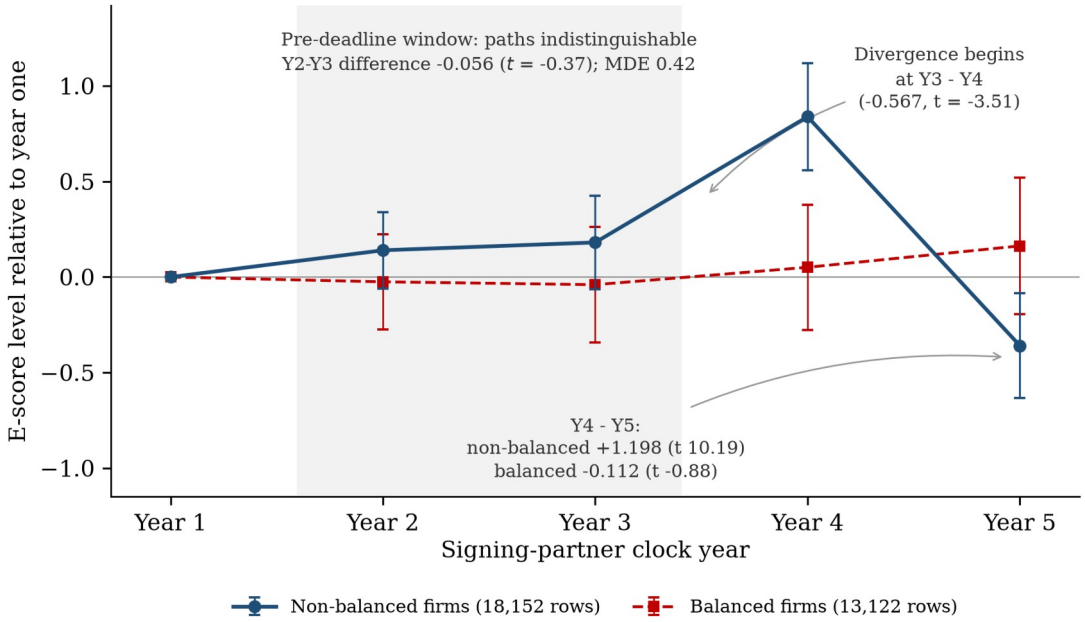


Figure 2. Pre-trend check by balanced status: regression-implied E levels against the year-one base. Non-balanced firms are those whose observed partner spells complete inside the window (18,152 rows); balanced firms are observed in all eleven years (13,122 rows). Points come from one firm-and-year fixed-effects regression with full controls (N = 31,274); the shaded band marks years two and three, the window before the deadline can bind.

Why is the non-balanced fall nearly twice the raw fall? Their spells finish inside the window, so the deadline arrives instead of being censored, and their year-five cells are joint cells more than twice as often; but the same selection also selects firms that rotate, and this design cannot tell the two apart.

4.5 The path is specific to the environmental pillar

Is this a tenure effect or a disclosure effect? Stacking the three pillars with firm-by-year, firm-by-pillar, and pillar-by-year fixed effects leaves identification to E's deviation from its sister pillars (Table 11, 93,822 pillar-firm-year cells). Run pillar by pillar, the fall is 0.646 in E (t = 7.44), +0.123 in S (t = 1.09) and -0.121 in G (t = -1.48).

Table 11. Pillar-specific path (stacked triple-difference: firm-by-year, firm-by-pillar, and pillar-by-year FE, controls, firm-clustered; 93,822 pillar-firm-year cells). The table reports the stacked E-specific contrast (+0.648, t = 7.45); the 0.646 (t = 7.44) quoted in the text is the same quantity estimated pillar by pillar instead, and the two agree to within rounding.

Quantity	Coefficient	SE	t
E-specific year 2	+0.058	0.080	0.73
E-specific year 3	+0.082	0.098	0.84
E-specific year 4	+0.513	0.108	4.75
E-specific year 5	-0.135	0.111	-1.21
E-specific year-four-to-year-five contrast	**+0.648**	**0.087**	**7.45**

4.6 Ownership

Splitting by state ownership (Wang, Wong, and Xia 2008), the fall is 0.822 (t = 7.76) among non-SOEs (20,295 firm-years) and 0.309 (t = 2.03) among SOEs (10,979), a difference of 0.51 (z = 2.75, computed from unrounded coefficients). Political evaluation and mandates leave less room for soft claims at state firms. Table 9 repeats the split at the pair-by-firm margin, where the two cells read +1.463 (t = 3.92) and +0.750 (t = 1.55); the magnitudes differ because the unit and the fixed-effect structure differ.

4.7 Realized penalties: null, and why

A rating response need not be a real-world response, so we test whether tenure predicts actual environmental penalties. The penalty models are null in every forward window: current year -0.0004 (p = 0.86; N = 19,391; 203 events), one year forward +0.0009 (p = 0.67; N = 16,317; 189 events), one-or-two years forward +0.0013 (p = 0.59; N = 16,398; 311 events). MDEs of 0.68, 0.72 and 0.61 percentage points against base rates of 1.05 to 1.90 percent bound the penalty response to be small in either direction.

A broader enforcement indicator looks better powered, so we explain why we do not use it. It carries 1,414 events against the register's 204 firm-years on the same 20,079-firm-year 2013-2020 window (7.04 against 1.02 percent; the 203 of the penalty models above is the current-year count inside the 19,391-row estimation sample, a subset of the same window - Appendix A.4), and on it the year-four-minus-year-five fall is +1.458 percentage points (t = +2.68), the first powered result in the section. But 73.7 percent of its events fall in 2017 and 2018, when the base rate runs

17.6 and 19.9 percent against 1.5 to 5.6 percent elsewhere, and dropping them flips the estimate to -0.462 (t = -0.98). A longer panel would not help either: 2021-2023 would shrink standard errors to 0.80 of their size and move the narrow-definition minimum detectable effect on the 2013-2020 window only from 0.66 to 0.53 percentage points (Appendix A.4; Note Section 8.23), a bound that sits between the 0.61 and 0.72 reported above.

One objection survives: the nulls could be truncation artifacts, because the extraction stops in 2020. We therefore rebuild the outcome from public announcements covering the whole window: 203 announcements from 119 firms, of which 124 firm-years fall on the primary sample (0.40 percent of firm-years). The year-four-to-year-five contrast is +0.142 percentage points (SE 0.132, t = +1.08; MDE 0.370), against +0.303 (SE 0.174, t = +1.75) on 2013-2020. The two measures agree on only 19 firm-years (correlation 0.13): disclosure duties bind on material penalties at the listed parent while the register also records subsidiary decisions, so we report both.

4.8 Annual-report text

Across the 762 environmental sections matched to an E score, E correlates with section length (r = 0.24) and quantitative density (r = 0.09) and negatively with commitment-word density (r = -0.22); the boilerplate ratio is orthogonal (r = 0.01). The firm-and-year contrasts use 779 sections from the 365 firms with more than one section. In firm-and-year fixed effects no clock-five contrast is distinguishable from zero: year five minus year four is -0.13 (t = -1.09) for length, -0.02 (t = -0.13) for quantitative density, +0.20 (t = +1.93) for commitment density and -0.18 (t = -0.91) for boilerplate, with minimum detectable effects of 0.29 to 0.55 standard deviations - 3.2 to 6.1 times the standardised rating fall of 0.09. The battery validates the outcome's construction but is uninformative about the fall.

4.9 Around the 2022 repeal

We read the pattern around the August 2022 repeal as before-and-after, not difference-in-differences: the cap applied to every listed client, so there is no never-treated group. Interacting year four and year five with a post-repeal indicator gives a

pre-repeal fall of +0.816 (t = 7.84) against +0.069 (t = 0.36) after, a difference of +0.747 (t = 3.26; MDE 0.64); window by window the fall is 0.755 (t = 6.76), 1.374 (t = 3.92) and -0.059 (t = -0.49) in 2013-2019, 2020-2021 and 2022-2023. The clock did not release: of 829 clients at five years in 2022, 56 percent kept a signing partner into 2023, 11 reached six, and the share of spells above five falls from 2.2 percent in 2018-2021 to 0.1 percent in 2025, using 2024-2025 audit records for this descriptive continuation check only and never for estimation. Post-2022 is therefore not an untreated regime, and we do not read the vanishing coefficient as evidence that the cap stopped binding. The obligation was continuous across the repeal: the CICPA Code of Ethics (2020) carried it from 2020 and the 2023 Measures (Article 13) restated it for listed companies, so we know of no year in our window in which the five-year cap was unenforced. We take no position on the channel.

4.10 Robustness

Table 12 collects the robustness estimates. Dropping every firm-change spell makes the fall stronger, not weaker (0.823, t = 7.69); only 104 of the 4,925 flagged mandatory changes rest on a prior clock below five, and removing those spells strengthens the fall, so clock misclassification works against us. The two-way pair-and-firm check (0.596, t = 5.02 on the 15,800 firm-years where a pair serves more than one client) is not firm-level averaging, and the two partial-replacement exclusions leave the fall unchanged (0.656 and 0.653).

Table 12. Robustness of the year-four-to-year-five contrast (firm and year FE, controls, firm-clustered unless noted). Every exclusion row starts from the 31,274-row primary sample and re-applies singleton absorption; flagged counts are identified on the clock-keyed sample, so N is not 31,274 minus the flagged count: the 1,282 name-matched years take 1,244 primary-sample rows with them (N = 30,030), the 493 non-conforming partial replacements take exactly 493 (N = 30,781), and the 728 partial replacements take 730 rows in total, the flagged rows plus rows of firms that become singletons (N = 30,544)

Specification	Fall	t	N
Primary (Table 4)	+0.646	7.44	31,274
Industry-by-year FE	+0.654	7.37	31,218
Two-way pair-and-firm FE (corroborating)	+0.596	5.02	15,800
Exclude whole spells containing an audit-firm change	+0.823	7.69	18,067
Exclude the 1,282 name-matched firm-change years	+0.726	8.11	30,030
Exclude the 493 non-conforming partial replacements	+0.656	7.41	30,781
Exclude all 728 partial replacements	+0.653	7.35	30,544
Exclude the ten largest pair-firm clusters (pair-by-firm)	+0.266	2.11	19,353

Specification	Fall	t	N
Balanced panel (firms observed all eleven years)	**-0.113**	**-0.89**	**13,122**
Pre-repeal window, 2013-2021	+0.816	7.84	23,597
Post-repeal window, 2022-2023	+0.069	0.36	7,677
Difference, pre-repeal minus post-repeal	+0.747	3.26	31,274

Two estimates carry the paper and both are pre-specified: the firm-level fall of Table 4 and the pair-firm joint fall of Table 8. Every split, subsample or window built around them is exploratory. The largest exploratory family, Table 9 - a family separate from the four firm-clock thresholds - has seven cells beyond the pre-specified E cell: a Bonferroni threshold of 0.05/7 gives |t| = 2.69, and only the non-state split (t = 3.92) clears it, while the S and G placebo pillars (|t| <= 1.32) and the state-owned split (t = 1.55) do not. The placebo family of four firm-clock thresholds (Appendix A.12) carries a Bonferroni threshold of |t| = 2.50 (0.05 over four tests): the four-year cell clears it (t = 2.82) and the six- and seven-year cells do not, so the graded pattern is a qualification of the title claim, not support for it.

4.11 The event study, and why it is not the anchor

Events are full pair replacements (Panel A, 15,451 events) or full changes out of a clock-five year (Panel B, 1,479 events); event -1 is omitted.

Table 13. Event-study cells (firm and year FE, controls, firm-clustered; omitted category event -1)

Event time	Panel A: all full changes (N = 27,149 cells, 4,035 firms)	Panel B: exact-five changes (N = 8,236 cells, 1,392 firms)
-4	-0.681 (t -1.52)	-0.565 (t -0.93)
-3	-0.097 (t -0.30)	-0.258 (t -0.55)
-2	+0.383 (t 1.69)	+0.250 (t 0.74)
0 (change year)	**-0.541 (t -3.56)**	-0.063 (t -0.38)
+1	-0.104 (t -0.59)	+0.167 (t 0.77)
+2	+0.292 (t 1.33)	+0.207 (t 0.63)
+5	+0.587 (t 1.80)	+0.409 (t 0.58)

Deposited file 13 carries the full -5 to +5 set.

Three honest readings. The pre-change leads are individually insignificant, but that is weak evidence of no anticipation: event -2 is tenure year four for most exact-five events, so the lead mechanically recovers the within-cycle peak. The only significant cell is the change year in Panel A (-0.541, t = -3.56), a rotation-year dip. Table 3's +0.003 (p = 0.98) for full changes out of a clock-five year and Table 13 Panel B's -0.063 for the exact-five change year are different specifications - a level contrast on the classification sample against an event-time cell - and are not expected

to coincide. And the cells are underpowered (MDE at event -2 in Panel B is 0.95 points against a primary fall of 0.646), so the primary evidence is Tables 4 and 8.

4.12 Bounding the disruption alternative

If any partner change depressed the rating through lost client-specific knowledge, the year-five fall would appear with no deadline at all. We bound that alternative with shocks that differ in whether they are anticipated, rebuilt from the signing-partner name sequence (2011-2023). A **deadline exit** occurs at individual tenure five or more (4,288 firm-years), an **early departure** below five with the partner stopping altogether (5,636), and a **reassignment** below five with the partner continuing elsewhere (9,822); we also use audit-firm changes (3,674 firm-years), a strictly larger rupture. Each enters with two leads and two lags, firm and year fixed effects, controls, firm-clustered, on 33,343 firm-years and 4,754 firms. Coefficients below are on E itself, so a positive number is a higher rating - the opposite of the contrast convention of Section 3.2; the same reversal applies to Table 13 and to Appendix Tables A.3 and A.4.

The disruption reading predicts a fall at each shock; it appears at none but the deadline. With the tenure path controlled, an early departure moves E by +0.003 (t = 0.03) with an MDE of 0.304 points, under half the primary fall, and its leads are jointly zero (chi-square(2) = 1.15, p = 0.56); an audit-firm change moves it by +0.096 (t = 0.64, MDE 0.419), also with leads jointly zero (chi-square(2) = 0.43, p = 0.81); the deadline exit moves it by -0.306 (t = -1.28). Two qualifications keep this from being a causal test: 92.1 percent of deadline exits occur at team clock five, so the indicator is largely a proxy for the clock (controlling the path cuts its t from -2.15 to -1.28), and the deadline-minus-early-departure difference in the event-year coefficient is -0.308 (t = -1.26), so the design separates deadline from turnover through the nulls on turnover.

What survives is a bound, not an estimate: turnover that is not the expiry of a clock does not move this rating by as much as 0.30 to 0.42 points, and the two shocks with clean pre-trends are the two that are null. That removes one reading of the fall; it does not identify the deadline's effect.

4.13 Matching on spell composition

The fall is estimated within firms, but spells that reach clock five could still be selected. Appendix A.11 matches at the spell level (treatment: reaching clock five; covariates: the ten controls plus the spell's opening-year E score): of 3,519 complete spells, 1,611 are treated and all match from 839 control spells. On the 8,984 matched firm-years the fall is 0.702 against 0.646, and inverse-probability weighting gives 0.652 - a 0.056-point span, about two-thirds of the primary standard error.

5. What the Evidence Can and Cannot Claim

Supported. The claims below rest on the two pre-specified estimates of Tables 4 and 8; every other contrast is exploratory (Section 4.10). Within firm-and-year fixed effects the E path along the partner clock is non-monotonic, with a year-four-to-year-five contrast of 0.646 ($t = 7.44$) that holds under industry-by-year fixed effects, under every measurable clock-misclassification margin, and under pair-by-firm fixed effects (0.295, $t = 2.27$; bootstrap $p = 0.03$). Among completing-spell firms, the joint-cluster differential is +0.653 ($t = 5.66$; $p < 0.001$) at the five-year point and +0.622 ($t = 6.13$) across the four-or-five window, against +0.057 ($t = 1.02$) outside it, and 1.20 on the joint subsample alone. Along the firm clock the fall is graded - +0.417 at four, +0.653 at five, not significant at six and seven (Appendix A.12) - not unique to five. It survives reweighting the joint firms' comparators on firm-level means (the joint-minus-firm-clock-below gap moves from +1.088 to +0.972, and to +0.673, $t = 2.21$, when persistence and spell-completion constraints are added, leaving 428 effective comparison firms (from 2,402; the 928-effective-firm figure of Table 10 row 9 belongs to the separate balanced-against-non-balanced battery of Section 4.4)) and spell-level matching, is specific to E and larger for non-state firms, and is zero in the two post-repeal years, when the clock still bound.

Not supported. No causal claim that the cap causes the fall: there is no control group, the design is a life-cycle contrast, and any shock moving with spell age within firm-year is differenced in only if orthogonal to the tenure dummies. The raw 0.646 is not the headline; pair-by-firm fixed effects cut it by more than half. The pattern does not survive the persistent-firm check, so selection into the joint cell stays a live alternative, and it is not detected in realized penalties or in annual-report text. It is not

turnover disruption: a partner leaving early and the whole firm being replaced both leave the rating unmoved, bounded at 0.30 and 0.42 points. The event study is underpowered. The outcome is one vendor's rating: the second E column in the public file is the same rating under a second reporting convention, not an independent rater, so no cross-rater replication is possible from the deposited materials. The limitation is not cosmetic - the Chinese vendors used in this literature disagree materially with one another and with international raters (Li and Cheng 2024) - so a second rater would not be expected to reproduce the level. And the five-year point is not unique at the pair-firm level: firm clock four carries +0.417 (t = 2.82), consistent with anticipation rather than single-year binding, which is why the four-or-five window is reported as a post hoc widening (Appendix A.12). Both nulls are bounds rather than refutations: the penalty designs detect 0.61 to 0.72 percentage points against base rates of 1.05 to 1.90 percent, and the textual battery detects 3.2 to 6.1 times the standardised rating fall of 0.09, so neither can exclude a real response of the size we estimate in the rating.

External validity. The joint result is established for firms whose partner spells complete inside the window - firms that enter or leave the panel, newly list, or change audit firms - and not for firms we can watch across the full window, where the path is flat. Whether that absence reflects a different response or censoring of the deadline we do not know; the 271-cluster joint subsample is marginally powered to say.

Benchmarks. The title joint contrast of 0.653 is 0.09 of the E standard deviation; the joint-subsample fall of 1.201 is 0.16; the all-cluster match-adjusted fall of 0.295 is 0.04.

6. Conclusion

China's five-year partner cap made the end of an audit relationship a known, dated event for nearly two decades. The rotation event has no average association with the environmental pillar; what is associated with it is the tenure path, which rises into year four and falls in year five. Under pair-by-firm fixed effects the fall survives where the audit-firm clock is in high tenure at the same time - a window with no statutory content, and one we cannot fully explain. It is specific to the environmental pillar and larger in non-state firms, and it is not detected among firms present in all eleven years, where the path is flat: the joint-cell fall is +0.044 (t = 0.19; minimum detectable effect 0.66) on the 1,203 firms we can watch for all eleven years against

+1.469 (t = 9.16) on the rest. That contrast is now detectable rather than suggestive: the firm-level cut detects 0.66 points, against 1.17 for the 271-cluster version. The joint result is established for firms whose partner spells complete inside the window, which is how the title should be read.

We move the rotation question from the event to the deadline: because the cap made the end date knowable, the informative margin is the final service year, where we find a concentration in the joint high-tenure window, not a broad tenure effect. Four numbers travel together and are not interchangeable: the raw firm-and-year fall of 0.646, the pooled pair-by-firm fall of 0.295, the joint-cluster interaction-implied contrast of 0.653, and the joint-subsample fall of 1.201; the post hoc four-or-five window (+0.622, t = 6.13) is a descriptive widening of the title estimand, not a fifth member of the set. The limits are plain: the design is associational, the fall is confined to firms whose partner spells complete inside the window, and a pre-trend check narrows that composition concern without removing it (-0.056, t = -0.37, through year three); firm-level entropy balancing on nine firm means moves the joint-cell fall only from +1.192 to +1.182, and adding a spell-completion constraint leaves it at +1.177 (Appendix A.13), so composition moves the estimate without overturning it. Whether the concentration reflects anticipation, firm review cycles, or selection is a question this design cannot settle; what it can do is locate the pattern precisely enough for a better design to test it. What travels is the location, and the location is measurable: the two clocks reach five together in 53.4 percent of clock-five firm-years among firms whose spells complete inside the window and 21.9 percent among firms we observe throughout, so the coincidence is not rare and can be looked for wherever a partner spell can be followed to its end. Two bounds rather than refutations: the turnover-shock designs bound disruption at minimum detectable effects of 0.30 and 0.42 points with estimated coefficients near zero, and the penalty and textual nulls bound a real-world response - 0.61 to 0.72 percentage points against base rates of 1.05 to 1.90 percent for penalties, and 3.2 to 6.1 times the standardised rating fall for text - so neither can exclude a response of the size we estimate in the rating.

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Appendix A: Data Construction and Sample Detail

Each subsection states the design and its estimates; the companion Data Availability Note (Sections 8.1-8.26) carries the full constructions, variable definitions and file-level provenance.

A.1 Sample funnel detail

Table 2's funnel in words: Branch A drops to 32,608 after excluding financial and ST firm-years, 32,537 after requiring the rotation classification and the clock, and 32,017 after requiring complete controls (4,571 firms); Branch B, which retains missing flags, ends its clock-keyed stage at 32,208 (4,666 firms); the tenure-1-5 and singleton-absorption screens of Table 2 then bring it to the 31,274-firm-year primary sample. The 191 firm-years between the branches all have a complete clock and controls; Section 8.7 of the companion Note reconciles both.

A.2 Clocks and transitions

The partner clock is defined in Section 3.1 and agrees with the within-firm individual clock in every firm-year where a name record exists. The firm clock is CSMAR's AuditorTenure, which resets on an audit-firm change (3,471 firm-years); both signing names change in 5,378 firm-years. Of the 24,696 within-spell transitions, 16,310 (66.0 percent) advance exactly one year, 297 (1.2 percent) skip a missing panel year, 1,951 (7.9 percent) are unchanged, 5,995 (24.3 percent) move backward and 143 (0.6 percent) jump forward by more than the calendar gap - an exact partition of the base. The non-advancing transitions are dominated by pair replacements (98.3 percent coincide with a signing-pair change), and the two partial-replacement exclusions leave the fall unchanged (0.656 and 0.653). Section 8.1 of the companion Note gives the full decomposition.

A.3 Rotation classification

A change is mandatory when the longer-serving outgoing partner has completed five consecutive service years. The flag marks 4,925 firm-years, of which 1,479 are full changes out of a clock-five year with the new pair at clock one. 104 flagged changes (2.1 percent) rest on a prior pair clock below five and fit unobserved pre-2013 service under Article 4; dropping every firm-change spell addresses this margin.

A.4 Penalty windows

The extraction file covers 2013-2020 (25,749 firm-year rows; 261 with a penalty). After listwise deletion and singleton absorption the estimation samples are 19,391 / 16,317 / 16,398 with 203 / 189 / 311 flagged cells and base rates 1.05 / 1.16 / 1.90 percent. The two-year window is mechanically truncated for 2019-2020 observations.

The merged panel carries a broader environmental-enforcement flag (1,414 events against the register's 204 on the same 20,079-firm-year window; base rates 7.04 and 1.02 percent) whose year-four-minus-year-five fall of +1.458 (SE 0.545, t = +2.68) we do not report as a finding (Section 4.7); the public-disclosure outcome rebuilt from cninfo announcements gives +0.142 (SE 0.132, t = +1.08) against +0.303

(SE 0.174, t = +1.75) on 2013-2020. Sections 8.4, 8.23 and 8.25 of the companion Note give the windows, the spike diagnostics and the retrieval screen.

A.5 Textual measures

Text-sample sizes and the four textual measures are described in Section 4.8; the retrieval funnel (1,119 PDFs, 969 usable sections, 762 matched) and the measure definitions are in Sections 8.10 and 8.11 of the companion Note.

A.6 Composition detail (Section 4.4)

Composition diagnostics are reported in Section 4.4 and Table 10; the balanced-panel cell counts, the four interaction coefficients, the implied levels and the joint-level subsample are in Section 8.17 of the companion Note.

A.7 Archive notes for replicators

(i) Cast stock codes to zero-padded strings before joining; the deposited package is type-consistent. (ii) The archived second-convention E_W copy equals E_H in all 10,352 overlapping rows; the raw extract, in which 32.7 percent of rows differ by rounding or revision noise, is deposited separately. (iii) PairID is the two-name signature as recorded (15,719 strings); sorting names within a pair would merge 1,914 order-variant pairs into 13,805. (iv) The 15,011 partner count covers the full 1997-2025 panel; the clock sample holds 9,628 names.

A.8 Entropy balancing of joint firms against the rest

Joint firms differ observably from the rest, so we re-estimate the joint fall after entropy balancing (Hainmueller and Xu 2013): treated are the 1,764 joint firms, controls the remaining 2,402, and the nine covariates are firm-level means (Table A.1). The optimiser converged (dual objective -0.207; weights 0.20 to 4.70).

Table A.1. Entropy-balancing covariate means

Covariate (firm mean)	Joint firms	Control, raw	Control, balanced	Raw gap	Balanced gap
State ownership	0.241	0.333	0.241	-0.092	0.000
Size	21.916	22.273	21.916	-0.356	0.000
Return on assets	0.044	0.037	0.044	+0.007	0.000
Growth	0.143	0.148	0.143	-0.005	0.000
Cash	0.245	0.205	0.245	+0.041	0.000
Leverage	0.367	0.406	0.367	-0.039	0.000
Age index	1.501	2.034	1.501	-0.533	0.000
Top-ten auditor	0.761	0.703	0.761	+0.058	0.000
Manufacturing (industry code C)	0.690	0.673	0.690	+0.017	0.000

On the entropy-balanced sample the joint fall is +1.182 (SE 0.135, t = 8.79) against +1.192 (t = 8.85) unweighted and the joint-minus-below gap narrows only from +1.088 to +0.972, so observable composition is not carrying the result.

A.9 Exploratory mechanism splits

Three splits ask whether the joint fall is larger where audit-firm review machinery should bind harder; all are exploratory and unadjusted for multiple testing. Auditor tier is a top-ten indicator from CSMAR; industry specialisation is the auditor's share of the firm's two-digit industry, split at the median (0.0796); client importance is the firm's assets as a share of the auditor's portfolio, split at the upper quartile (0.0100).

Table A.2. Exploratory mechanism splits (joint-cell fall by arm)

Split	Joint fall, high arm	t	Joint fall, low arm	t	Difference	t
Top-ten auditor	+0.859	4.45	+1.365	4.48	-0.506	-1.57
Industry specialist	+0.898	3.97	+1.102	4.75	-0.204	-0.71
Top-quartile client importance	+0.416	1.06	+1.111	5.97	-0.695	-1.71

The fall appears in both arms of every split and is if anything smaller where the review story predicts it larger; the client-importance arm is underpowered (MDE about 1.1 points), so its smaller estimate is not evidence of absence.

A.10 Turnover-shock construction and cells (Section 4.12)

Partner identities are the signing-partner name strings in the audit panel, de-duplicated within firm-year: 119,737 firm-partner-year records covering 12,015 partners over 2011-2023. An individual tenure clock counts consecutive years in which the same name signs the same client; an exit is the last year of that run, kept when the client is observed the following year (36,448 exits). A deadline exit occurs at individual tenure five or more; an early exit is a reassignment if the same name signs another client next year and a departure if not. Panel B, which adds the tenure path, is

the specification quoted in Section 4.12; Section 8.21 of the companion Note gives the construction.

Table A.3. Panel A: each shock estimated separately (t statistics in parentheses; coefficients are on E itself, so a positive number is a higher rating - the opposite of the contrast convention of Section 3.2)

Event time	Deadline exit	Early departure	Reassignment	Audit-firm change
Firm-year events	4,288	5,636	9,822	3,674
k = -2 (two years before)	+0.128 (1.14)	-0.050 (-0.49)	-0.104 (-1.20)	-0.092 (-0.73)
k = -1 (one year before)	**+0.489 (3.79)**	+0.015 (0.14)	-0.125 (-1.41)	-0.063 (-0.47)
k = 0 (event year)	**-0.286 (-2.15)**	-0.011 (-0.11)	+0.009 (0.10)	-0.027 (-0.20)
k = +1 (one year after)	+0.036 (0.27)	+0.041 (0.38)	-0.169 (-1.94)	+0.080 (0.56)
k = +2 (two years after)	+0.129 (0.99)	+0.143 (1.31)	-0.037 (-0.43)	+0.043 (0.34)

Table A.4. Panel B: three shocks jointly, tenure path controlled (t statistics in parentheses; coefficients are on E itself, so a positive number is a higher rating - the opposite of the contrast convention of Section 3.2)

Event time	Deadline exit	Early departure	Audit-firm change
k = -2 (two years before)	+0.138 (1.08)	-0.039 (-0.38)	-0.090 (-0.70)
k = -1 (one year before)	+0.257 (1.62)	+0.076 (0.70)	-0.026 (-0.19)
k = 0 (event year)	-0.306 (-1.28)	+0.003 (0.03)	+0.096 (0.64)
k = +1 (one year after)	+0.133 (0.96)	+0.123 (1.09)	+0.138 (0.93)
k = +2 (two years after)	+0.179 (1.37)	+0.195 (1.72)	+0.080 (0.61)

Reassignment enters Panel A but is not repeated in Panel B. Reassignment and early departure are the two halves of the below-five exit margin: a partner who leaves before completing five consecutive years is reassigned if the same name signs another client in the following year and departs if it does not. Panel B therefore enters the deadline exit together with one below-five disruption (early departure) and one firm-level rupture (audit-firm change), which are the three shocks Section 4.12 uses to bound the disruption reading; the complementary half is reported on its own window in Panel A.

The deadline shock's significant one-year-ahead lead in Panel A is the tenure path, not anticipation: 92.1 percent of deadline-exit firm-years sit at team clock five and 83.5 percent of the firm-years one year before sit at team clock four, so the lead recovers the year-four peak of +0.490. The deadline-minus-early-departure difference in the event-year coefficient is -0.279 (t = -1.65) in Panel A and -0.308 (t = -1.26) in Panel B; as in Section 4.12 these are coefficients on E, where positive is a higher rating.

A.11 Propensity-score matching of tenure spells (Section 4.13)

Table A.5. Spell-level covariate balance (opening year)

Covariate	Treated	Control, raw	Control, matched	SMD, raw	SMD, matched
Size	21.779	22.023	21.788	-0.192	-0.007
Return on assets	0.047	0.034	0.048	+0.204	-0.028
Leverage	0.368	0.402	0.359	-0.155	+0.045
State ownership	0.384	0.437	0.384	-0.108	+0.001
Age index	1.342	1.735	1.307	-0.355	+0.032
E score	60.398	60.088	60.549	+0.044	-0.022
Propensity score	0.518	0.407	0.518	+0.712	+0.000

Spell-level matching is described in Section 4.13; the propensity model and covariate balance are in Table A.5 and Section 8.22 of the companion Note. On the 8,984 matched firm-years the fall is 0.702 against 0.646; the MDE on the year-five level rises to 0.525, so that level alone is not distinguishable from zero, but the fall is not. Inverse-probability weighting gives 0.652.

A.12 Placebo joint thresholds and cluster-level reweighting (Sections 4.3 and 4.10)

Does the firm clock have to read five? A sharper test holds the partner clock at its deadline year and re-cuts the cell on the audit-firm clock the firm carries in its first partner-clock-five year, re-estimating the fall inside each cell with the Table 6 partition regression; the benchmark reproduces the joint cell exactly (+1.192, t = 8.85).

Table A.6. Firm-clock placebo: the year-four-to-year-five fall by the audit-firm clock at the partner's clock-five year (firm and year FE, controls, firm-clustered; positive = decline)

Audit-firm clock at the partner clock-five year	Fall	SE	t	Firms	MDE (80%)
4	+0.423	0.324	1.31	128	0.91
5 (joint cell)	**+1.192**	**0.135**	**8.85**	**1,764**	**0.38**
4 or 5 (joint high tenure)	**+1.114**	**0.126**	**8.85**	**1,892**	**0.35**
6	+0.215	0.357	0.60	173	1.00
7	+0.228	0.304	0.75	157	0.85
All other values	+0.170	0.145	1.17	1,944	0.41

Only the five-year cell moves. Each placebo cell is indistinguishable from zero and significantly below the joint cell: joint minus firm clock 4 is +0.769 (t = 2.19), minus 6 +0.976 (t = 2.55), minus 7 +0.964 (t = 2.90), minus all other values +1.022 (t = 5.16). The cells are thin (128 to 173 firms; MDEs near one point). Pooling the four and five cells gives +1.114 (SE 0.126, t = 8.85) on 1,892 firms, against -0.020 (t = -0.08) below four and +0.191 (t = 1.33) above five.

The same placebo at the pair-by-firm level is graded, not knife-edged. Redefining the coincidence at firm clock 4, 5, 6 and 7 in the Table 8 design returns +0.417 (SE 0.148, t = 2.82; 137 clusters), +0.653 (SE 0.115, t = 5.66; 970), +0.056 (SE 0.263, t = 0.21; 157) and -0.216 (SE 0.177, t = -1.22; 159). The firm-clock-four estimate is not contamination (1.5 percent of those clusters also hold a joint cell; restricting to clusters with none raises it to +0.457, t = 2.96) and not a single-year artifact (dropping each year leaves +0.362 to +0.515, t from 1.87 to 3.09), but it fades when clusters must hold three cells (+0.406, t = 1.79; 76 clusters). A dated deadline should not be expected to bind in a single year, and the four-year reading is consistent with adjustment beginning in the penultimate year of the association; we report it rather than build on it. The two-year window gives an implied joint effect of +0.622 (SE 0.102, t = 6.13; 1,105 clusters), with a leave-one-year-out range of +0.368 to +0.815 and +0.621 (t = 6.05) after dropping the ten largest clusters.

Cluster-level reweighting. The largest standardized difference between the 970 joint clusters and the 6,781 others is 0.613 (listing age) across twelve cluster means. Entropy balancing reweights the non-joint clusters to match the joint clusters exactly on all twelve (dual objective -0.540; row weights 0.04 to 15.34). The Table 8 interaction then gives D2 = +0.140 (t = 1.63), D2 x joint = +0.608 (t = 4.13) and an implied joint effect of +0.748 (SE 0.120, t = 6.23) against +0.653 (SE 0.115, t = 5.66) unweighted, identical to the Table 8 cell. The gap that carries the result goes from +0.596 to +0.608, and to +0.837 (t = 4.12) when the share of cells at clock four or five is added - the dimension on which joint clusters differ most, +2.13 standard deviations.

A.13 Firm-level symmetric balance and persistence-constrained reweighting (Section 4.4)

The cluster-level reweighting of Appendix A.12 has a firm-level counterpart, run on the same estimation sample. The tenure-one-to-five and singleton screens of Table 2 reconstruct that sample exactly from the deposited main panel (32,208 rows to 31,734 to 31,274 firm-years; 4,166 firms; 2,018 joint firm-years), and the Table 6 benchmark reproduces to three decimals (+1.192, t = 8.85, in the joint cell), so the pipeline below is validated against the published cells before any new constraint is added. Each firm is assigned to the Table 6 cell of its first partner-clock-five year, and firm-level covariates are means over the firm's observed years. The treated group is the 1,764 joint firms; the comparison group is the remaining 2,402.

Table A.7 reports the raw balance, which the manuscript previously showed only in balanced form. The largest imbalance among the twelve covariates is listing age (-0.63 standard deviations); the persistence descriptors are nearly as large (years observed -0.51, years covered -0.51, presence in all eleven years -0.53), and two quantities are definitional rather than balanceable: every joint firm's partner spell reaches clock five inside the window (1.00 against 0.61), and 44.7 percent of a joint firm's observed years sit at partner clock four or five against 29.4 percent for the rest.

Table A.7. Firm-level balance of joint firms against the rest (firm means over observed years; standardized differences on the pooled standard deviation; 1,764 joint firms against 2,402 others)

Covariate (firm mean)	Joint firms	Other firms	Std. diff.	t
State ownership	0.242	0.333	-0.203	-6.53

Covariate (firm mean)	Joint firms	Other firms	Std. diff.	t
Size	21.916	22.273	-0.306	-9.83
Return on assets	0.044	0.037	+0.154	+4.92
Growth	0.143	0.148	-0.030	-0.97
Cash	0.245	0.205	+0.141	+4.43
Leverage	0.367	0.406	-0.216	-6.90
Listing age (index)	1.501	2.034	-0.630	-20.14
Female directors	0.057	0.053	+0.017	+0.52
Executive education	2.772	2.644	+0.089	+2.85
Pay gap	6.568	6.998	-0.092	-2.97
Top-ten auditor	0.761	0.703	+0.149	+4.77
Manufacturing	0.693	0.694	-0.002	-0.08
Years observed	6.58	8.18	-0.506	-16.15
Years covered	6.92	8.60	-0.514	-16.38
Observed all eleven years	0.158	0.385	-0.527	-17.17
Spell reaches clock five	1.000	0.607	+1.137	+39.39
Years at clock four or five	0.447	0.294	+0.811	+25.88

Table A.8 reports the reweighting. Entropy balancing (Hainmueller and Xu 2013) reweights the 2,402 comparison firms onto the 1,764 joint firms; row 2 reproduces the Appendix A.8 constraint set (the nine firm means), row 3 adds the two persistence descriptors, and row 4 adds the spell-completion indicator - the firm-level analogue of the clock-four-or-five cell share that Appendix A.12 adds at the cluster level, and the dimension the composition concern is about. All three sets balance to machine precision (largest absolute standardized difference below 1.5e-14). Adding persistence costs almost nothing (effective comparison firms 1,634 to 1,629); adding completion drives the 943 firms with no clock-five year inside the window to zero weight and leaves 428 effective comparison firms, with weights from 0 to 21.7. The Table 6 specification is then re-estimated with the firm weights applied to firm-years. The joint fall is +1.182 (t = 8.79) under the Appendix A.8 constraints, +1.182 (t = 8.78) with persistence added, and +1.177 (t = 8.74) with completion added; the joint-minus-below gap moves from +1.088 unweighted to +0.972, +0.993 and +0.673 (t = 2.21). The last two rows restrict the Table 6 specification by panel persistence: the joint-cell fall is +0.044 (t = 0.19; MDE 0.66) among the 1,203 persistent firms (279 in the joint cell) against +1.469 (t = 9.16; MDE 0.45) among the rest (2,963 firms; 1,485 in the joint cell), so the cell-level pattern is a non-persistent-firm phenomenon even though the differential survives the reweighting.

Table A.8. Persistence-constrained reweighting and the joint-cell fall by persistence (firm and year FE, controls, firm-clustered; positive = decline). Rows 2-4 apply entropy-balancing weights at the firm level to the Table 6 specification; the effective-comparison column counts comparison firms after weighting (Kish effective sample size). The last two rows restrict the same specification by panel persistence.

Specification	Joint fall	SE	t	Below-five fall	Joint-minus-below gap	t	Effective comparison firms
Unweighted (Table 6 benchmark)	+1.192	0.135	8.85	+0.104	+1.088	4.54	2,402
Nine firm means	+1.182	0.135	8.79	+0.210	+0.972	3.74	1,634

Specification	Joint fall	SE	t	Below-five fall	Joint-minus-below gap	t	Effective comparison firms
(Appendix A.8 set)							
Plus years observed and covered	+1.182	0.135	8.78	+0.189	+0.993	3.70	1,629
Plus spell completion	+1.177	0.135	8.74	+0.504	+0.673	2.21	428
Persistent firms only (1,203 firms; 279 joint-cell)	+0.044	0.235	0.19	-0.407	+0.451	1.32	-
Non-persistent firms (2,963 firms; 1,485 joint-cell)	+1.469	0.160	9.16	+0.816	+0.652	1.80	-

Data Availability

The findings draw on third-party licensed data and on public data; no human-subject or confidential personal data are used. Audit, financial, executive and environmental-penalty records come from CSMAR, available only under a commercial license and not redistributable. The Huazheng ESG pillar scores are public through Mendeley Data (Lin 2025, DOI: 10.17632/8fh8wtn6p5.2); annual-report text comes from public exchange disclosures. Under the Journal of Accounting Research Data and Code Sharing Policy, every variable, the sample funnel, and the estimation scripts are described in Section 3, Appendix A and the companion Data Availability Note. The study-specific code, the constructed identifiers, and the non-licensed derived data will be deposited at the repository designated by the editors upon acceptance; the extract and code are available to the editor and reviewers on request, subject to CSMAR licensing.